How SIMT is modeled without pretending lanes are CPU threads

Lane masks stay inside a warp; scheduling interleaves resident warps across SMSP issue partitions.

CPU-style thread interleaving

host threads own independent PCs

Tradeoff

accurate per-thread control; high host/event cost

SIMT warp lockstep

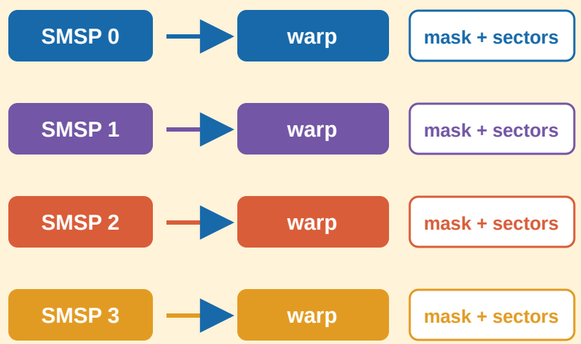
one warp instruction, many lanes

Tradeoff

excellent lane efficiency; divergence and masks matter

Our native replay

recorded warp streams + masks



Our tradeoff

one host actor per SM; lanes stay compact and auditable

Key distinction **we do not spawn 32 host threads per warp**

the trace supplies lane masks/addresses; sim-sm interleaves resident warps, and sim-mem aggregates sectors at cache acceptance

Caption. SIMT modeling is a hierarchy: lanes execute one warp instruction together, while resident warps interleave at SMSP issue boundaries. The replay preserves the lane evidence without paying CPU-style per-thread event cost.

Source: sim/sm/src/issue.rs, sim/sm/src/replay.rs, static SASS masks, placement metadata, and custom SIMT differential evidence.